ERA Official Demo: Stage-1 Reproduction Report

Reproduction Experiment Log

June 29, 2026

1 Overview

This report documents a reproduction of the **official minimal demo** shipped with `google-research/era`, the open-source code for the Nature paper *An AI system to help scientists write expert-level empirical software.*

An important clarification: this is **not** the full scRNA-seq or COVID scientific experiment from the paper. It is the official minimal, runnable demo included in the repository. Its purpose is to first get the entire ERA pipeline working locally end-to-end (LLM sampling, sandboxed execution, tree search, scoring, and iterative improvement), confirming that the system runs correctly and exhibits the expected "gradual self-improvement" behaviour, as a Stage-1 foundation for the larger experiments to follow.

2 The Task

The demo is a Kaggle / California Housing–style **regression task**: given housing-related features, predict a continuous target value. ERA first evaluates an initial baseline solution and then runs a tree search that repeatedly generates, executes, and improves candidate programs.

3 The Score

The scoring metric is the **negative root-mean-squared error (negative RMSE)**, i.e. $-$RMSE. Because of the negative sign, a **higher score (closer to** 0**) means a better model**.

4 Key Results

The core results of this reproduction are summarised below.

Stage	**Best score (negative RMSE, higher is better)**
Initial baseline	-0.7339448894714724
10-iteration best	≈ -0.5785
30-iteration best	-0.5775746202235905

Table 1: Best score of the ERA official demo at each stage.

5 Search Progress

Figure 1 shows the best-so-far score as a function of the search step during ERA tree search. Step 0 is the initial linear-regression baseline; a large jump occurs immediately at step 1, after which the search makes small, steady improvements on gradient-boosting / feature-engineering solutions before quickly reaching a plateau.

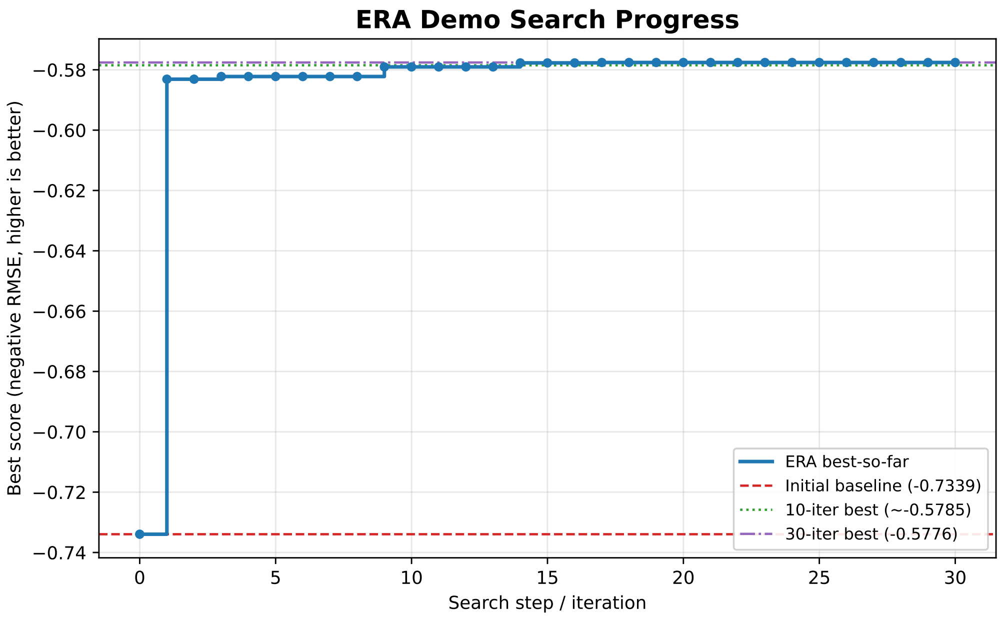

Figure 1: ERA demo search progress: best-so-far score (negative RMSE) versus search step. The red dashed line is the initial baseline, the green dotted line is the 10-iteration best, and the purple dash-dotted line is the 30-iteration best.

6 Interpretation

- **The ERA main loop was successfully reproduced.** The system completes the full closed loop of sample → sandboxed execution → scoring → selection and iteration, and exhibits the expected self-improvement behaviour.
- **Solution quality improved substantially.** The search started from a simple linear-regression baseline (score ≈ -0.7339) and automatically evolved into a solution centred on gradient boosting (`HistGradientBoostingRegressor`), with extensive feature engineering (including location/distance features) and an ensemble of three gradient-boosting models (score ≈ -0.5776).
- **It plateaus quickly.** Thirty iterations improved only marginally over ten iterations (roughly $-0.5785 \rightarrow -0.5776$), indicating that the search space of this toy task is small and that ERA approaches the plateau quickly, with rapidly diminishing returns from further iterations.

7 Next Step

The next experiment is to compare, on the same task, **ERA tree search** against **best-of-N independent LLM sampling**. Under an equal LLM-call budget, we will evaluate the final score and sample efficiency of both approaches, in order to measure the actual gain of ERA's "tree search + iterative improvement" over plain independent parallel sampling.